

ARTICLE

Utilizing bacteriophage to define the minimum residence time within a plug flow reactor

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Abstract

As part of a viral mitigating strategy for continuous bioprocessing, that utilizes a plug flow reactor (PFR) for continuous viral inactivation (CVI), understanding the minimum residence time as a function of reactor scale and operational conditions is critical. An empirical-based model was utilized to calculate the minimum duration a virus particle experiences within a plug flow reactor as a function of reactor design and operational conditions. This empirical model's calculations were challenged by pulse injecting the bacteriophage Φ X-174 in non-inactivating conditions and monitoring the discharge of the PFR with infectivity assays. The initial proposed empirical model, with the constraint of requiring an operational Dean number of >100 , proved to be effective at calculating first breakthrough of Φ X-174 but only for the appropriate Dean number conditions. With the knowledge gained from the first empirical model, a second was generated to eliminate the Dean number constraint. This second modified empirical model proved to be successful at calculating the first breakthrough at all Dean number's tested, however CVI operation at the lower Dean's number will lead to an increased asymmetry (i.e., increased tailing) in the residence time distribution.

KEYWORDS

continuous manufacturing, Dean vortices, plug flow reactor, scale down, scale up, viral inactivation

1 | INTRODUCTION

As continuous bioprocessing approaches GMP implementation and the opportunity to add to the global drug supply manufacturing process, the establishment of a robust scalable continuous viral inactivation (CVI) solution is vital (Klutz et al., 2015). There are essentially two viable options for post protein A, low pH CVI. The first being an operational modification of the traditional tank-based paradigm in which multiple tanks are operated periodically such that while one vessel collects the product the others perform the acidification, incubation, and subsequent neutralization (i.e., Pall Cadence™ Virus Inactivation System or a three tank rotational system (Johnson et al., 2017). The second utilizes a plug flow reactor (PFR), such as a tubular reactor or column, in which the product

reaches a targeted pH set point before entering a flow path. The product is subsequently held in the PFR as a function of the reactor volume and volumetric flow rate (Arnold et al., 2019; Gillespie et al., 2018; Klutz et al., 2016).

While a multi-tank system has the appeal of being similar to traditional batch operation and the appearance of being fundamentally simpler, a PFR-based CVI affords the unique benefits that is more compatible to accommodate the challenges associated with continuous bioprocessing. A PFR allows for the hydraulic linking of unit operations without the need for surge vessels. This provides the advantages of efficient chemical sanitization of the product contacting surfaces of the downstream system between Protein A cycles, and allows for the independent processing of each cycle such that there is no blending of product from cycle to cycle while the

product remains in the downstream chain of unit operations. With these advantages comes a new challenge in the form of assuring comparable incubation duration of the product stream of tank-based operation to a PFR due to dispersive effects.

Multiple publications have been authored by Boehringer Ingelheim on the topic of dispersive affects within a PFR (Amarikwa et al., 2019; Brown et al., 2020; Orozco et al., 2017; Parker et al., 2018), specifically the Jig-in-a-Box (JIB) design, which decreases dispersion via Dean vortices. The Dean number is defined by Equation (1), where ρ is the fluid density, u is average linear flow velocity, D is flow path internal diameter, μ is dynamic viscosity, and R_c is the radius of curvature of the serpentine path. The central focus of these publications has always been on the relationship between the dynamic fluid mechanics elicited by Dean vortices, and the residence time distribution within the PFR. To control duration, the volume of the reactor, flow rate, and dispersive effects need to be taken into consideration.

Previous work had shown that the residence time distribution (RTD) of a pulse tracer within the JIB changed as a function of Dean number (De), internal diameter (D), and reactor path length (L). This behavior of the RTD becomes relevant during the scale up/down process as both the flow rate and the reactor design (i.e., internal diameter and/or the reactor path length) needs to be adjusted to accommodate scale change. The pulse tracers were analyzed by fitting a Gaussian distribution to the exiting peak and calculating a variance (σ_{time}^2) value. The data generated by these experiments, shown in Figure 1 (Brown et al., 2020), were normalized by Equations (2) and (3) where HETP is height equivalent to a theoretical plate, T_{Ave} is the mean residence time, Q is the flow rate, and RV is the reactor volume. This data could then be used to either extrapolate HETP data to other internal diameters where d is reduced HETP, or to create an empirical model to calculate the expected variance as a function of Dean number (Equation 4). To calculate when the least incubated fluid element would first exit the reactor (i.e., minimum residence time or T_{min}), Equations (1)–(4) are combined to determine a variance value, and the root value taken to produce a standard deviation (σ_{time}). The final calculated T_{min} value, $T_{min}(5\sigma)$ is produced by Equation (5), in which five standard deviation values are subtracted from the average residence time for a given path length and Dean number combination. Justification for the use of five standard deviations for T_{min} is discussed in the publication (Brown et al., 2020). Equations (1)–(5) can be combined to create a simplified version (Equation 6). This empirical model was thought only to be valid for a defined set of conditions (i.e., Dean number ≥ 100) due to the nonuniform behavior of the exiting pulse injections at lower Dean numbers. Modifications to the empirical model to extend $T_{min}(5\sigma)$ to Dean number < 100 will be covered in Sections 3:

$$De = \left(\frac{\rho u D}{\mu} \right) \sqrt{\frac{D}{2R_c}}, \quad (1)$$

$$HETP = \frac{\sigma_{time}^2 L}{T_{Ave}^2}, \quad (2)$$

$$T_{Ave} = \frac{RV}{Q}, \quad (3)$$

$$h = \frac{HETP}{D} = f(De), \quad (4)$$

$$T_{min}(5\sigma) = T_{Ave} - (5\sigma_{time}), \quad (5)$$

$$T_{min}(5\sigma) = T_{Ave} - 5 \sqrt{\frac{f(De) D T_{Ave}^2}{L}}. \quad (6)$$

To experimentally quantify T_{min} , multiple types of materials (i.e., monoclonal antibodies, riboflavin, gold nanoparticles, and dextrose) and detection methodologies (i.e., UV-Vis absorbance and total organic carbon measurement) (Amarikwa et al., 2019) were used. While each testing chemical expanded the understanding of the residence time within the PFR, they were not infectious virus particles. This study will focus on the application of the minimum residence time by conducting targeted testing with a surrogate viral particle (i.e., bacteriophage Φ X-174) quantified with an infectivity assay, with chosen parameters selected based on learnings from Brown et al. (2020).

2 | MATERIALS AND METHODS

2.1 | Mobile phases and flow chamber

The JIB design was created from previous development projects (Amarikwa et al., 2019; Brown et al., 2020; Orozco et al., 2017; Parker et al., 2018) and was 3D printed utilizing SLA Technology by 3D Systems (Rock Hill, SC). For this experimental series, a 0.318 cm and 0.635 cm internal diameter JIB design were used. The riboflavin, Tris-buffered saline (TBS), sodium chloride and dextrose used in creating the mobile phases were purchased through Thermo Fisher Scientific. The viscosities of the solutions were determined by a

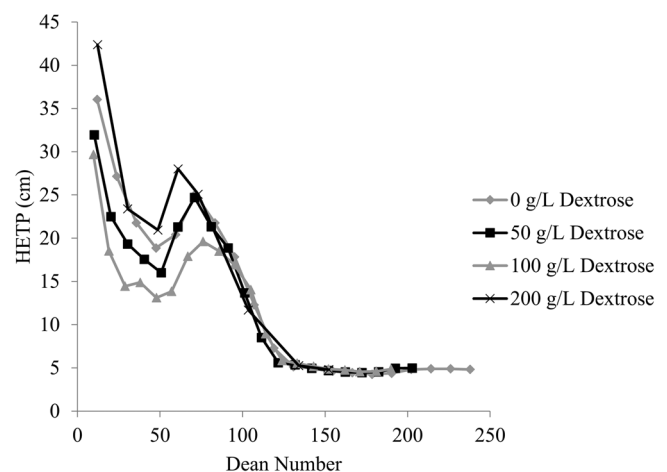


FIGURE 1 The influence of flow rate and viscosity on the peak spreading of the exiting pulse injections, and the nonuniform behavior of experiments conducted at Dean number ≤ 100 . Recreated from Brown et al. (2020). HETP, height equivalent to a theoretical plate

microVISC S Viscometer utilizing an A05 Chip. The densities of the solutions were determined by a calibrated pipette and a scale.

2.2 | Preliminary proof of concept (NaCl pulse)

To illustrate the $T_{\min}$ phenomenon experimentally before the use of Φ X-174, a JIB was first flushed with high purity water such that the conductivity of the JIB outlet was below the limit of detection of the Akta Avant 150's conductivity probe resulting in a reading of 0 mS/cm. Next, a pulse injection of 150 mM NaCl (3% v/v) was introduced into the JIB and flushed with high purity water at a constant volumetric flow rate spanning 5–100 ml/min in 5 ml/min increments. These experiments were carried out in the labs of Boehringer Ingelheim in Fremont, California.

2.3 | Bacteriophage selection

Φ X-174 and the corresponding host bacteria *E. Coli* C were purchased from ATCC (ATCC Catalog #: 13706-B1 and 13706, respectively). Φ X-174 was propagated, purified and quantified by standard plaque-forming assays as described in Lute et al. (2007).

The bacteriophage Φ X-174 was chosen as the appropriate tracer for this experiment since it is similar in size and shape to xenotropic murine leukemia virus (X-MuLV), the virus typically used in viral clearance studies. Both are relatively spherical and Φ X-174's hydrodynamic diameter is 30–34 nm while X-MuLV is 80–20 nm. This difference in size is not expected to impact viral particle residence time based on previous work that incorporated pulse injection experiments with a wider range of hydrodynamic diameters with no observable impact (Amarikwa et al., 2019). In addition to these similarities, Φ X-174 has some innate characteristics that make it an optimal testing bacteriophage. Φ X-174 is a relatively resilient bacteriophage where chances of loss in infectivity while suspended in an appropriate mobile phase conditions are low, but can also be easily sanitized with 0.1 M NaOH and a reasonable contact time. The surface characteristics of this bacteriophage are relatively inert compared to other viruses. Previous experience with this bacteriophage had shown significantly less surface adsorption relative to other viral models to both positively charged, hydrophobic, and multi-modal chromatography resins (Brown et al., 2017). Therefore, the probability of the virus nonspecifically adsorbing in a slightly basic solution with a low ionic strength (i.e., pH 7.5 with 150 mM NaCl) to the 3D printed plastics was low making Φ X-174 an optimal worst case for minimal residence time. The plaque morphology of Φ X-174 was also advantageous. Φ X-174's plaque forming units (pfu) create large, bullseye type plaques that are easy to identify.

2.4 | Φ X-174 pre-experiment workup

All Φ X-174 containing experiments were carried out in the labs of Keck Graduate Institute (KGI) in Claremont, California. First, to

determine the preservation of infectivity of Φ X-174 flowing through the JIB, a pulse injection of Φ X-174 was introduced into the JIB at the highest Dean number (i.e., high flow rate and low viscosity) which corresponds to the most chaotic flow stream due to Dean vortices. The discharge of the JIB was then collected and titered. The volume adjusted pfu levels of the initial injection matched the JIB discharge within the typical error associated with a titrating assay (i.e., ± 0.5 logs). A sanitization program was then created to thoroughly sanitize the injection valve, JIB, and outlet valve with 0.1 M NaOH with a contact time of ≥ 15 min. After the sanitization cycle, no infectious particle remained in the outlet lines.

2.5 | Determining minimum residence time (Φ X-174)

A 0.318 and 0.635 cm i.d. 3D printed JIBs were tested using an AKTA™ Explorer 100 by GE Healthcare. To prepare for the experiment, a 30 ml aliquot of the mobile phase was taken and set aside for spiking. The aliquot was then spiked at 0.06% (v/v) targeting a mobile phase concentration of $10^{6.5}$ pfu/ml. The spike was purposely spiked at a considerably low level to ensure the fluid properties of the experimental injection were not changed by the Φ X-174 spike. Using a syringe, 25 ml of spiked sample was then loaded into a 50 ml capacity Superloop by GE Healthcare while the remaining sample was held on the bench as a hold sample to determine if significant Φ X-174 inactivation occurred as a function of mobile phase condition independent of flowing through the JIB. Φ X-174 never experienced an off target mobile phase concentration within the typical error of a titrating assay (i.e., ± 0.5 logs). The tare weight of the empty fraction containers were determined before the experimental run.

To start the experiment, the Akta began pushing mobile phase through the injection valve, in the "Inject" position, into the Superloop to introduce 15.6 ml of the Φ X-174 spiked buffer into the JIB where the discharge was directed into a large volume container. The injection valve then switched to "Load" position to stop flow of mobile phase through the Superloop, and redirected flow to go directly to the JIB to flush out the injection with the discharge remaining in the same large volume container. The internal diameters, number of JIBs connected in series, flow rates, and mobile phases tested in this study are outlined in Table 1. To increase the viscosity of the TBS mobile phase, dextrose was added (i.e. 200 g/L) to simulate the kinematic viscosity of a mAb solution of approximately 50 g/L.

After a predetermined amount of time, the outlet valve switched to direct flow from the large volume container to then begin collecting three fixed volume fractions. When the experiment was completed, the three small and one large container were weighed. To sample for Φ X-174, the liquid held up in each of the capillary tubes of the outlet valve were drained for a sample of $\sim 100 \mu\text{l}$. Each sample was then titered separately without dilution. The sample left behind in the capillary tube would have been the last drop of that fraction and can be thought of as an instantaneous grab sample.

Internal diameter (cm)	Volume (ml)	Flow rate (ml/min)	Viscosity (cP)	Replicates
0.318	125	12.5, 25, 37.5	1.03	2
0.635	520	25, 50, 75	1.03, 1.71	2
0.635	3120	50	1.03, 1.71	2

TABLE 1 Viscosities, reactor volumes, and flow rate combinations tested with Φ X-174

After all post-Akta experiment activities were completed, the remaining spiked mobile phase was expelled from Superloop, and the Superloop was taken offline. The injection loop position was then replaced with PEEK capillary tubing and the sanitization step outlined above was completed, and then subsequently quenched with TBS.

3 | RESULTS AND DISCUSSION

3.1 | Preliminary work—NaCl pulse injection

For a visualization of $T_{\min}$, one 0.635 cm i.d. JIB was first flushed with high purity water such that the conductivity probe of the Akta read 0 mS/cm, subsequently a pulse injection of 150 mM NaCl (3% v/v) was introduced into the JIB and flushed with high purity water. Figure 2a,b displays the results of the experiments spanning 5–100 ml/min in 5 ml/min increments. $V_{\min_apparent}$ (Figure 2a) corresponds to the volume of water flushed before the Akta's conductivity probe detected the first nonzero data point (~ 0.003 mS/cm). This volume is larger than the actual $V_{\min}$ as the conductivity probe, due to limit of detection, would not detect the actual instance of first breakthrough of the salt. As the volumetric flow rate increases, the Dean number also increases. A higher Dean number corresponds in an increase in the chaotic nature of Dean vortices decreasing dispersion resulting in a larger volume of flushing to observe conductivity breakthrough. This is seen in Figure 2a where for the slowest flow rate, the first instance of conductivity

breakthrough occurs at $\sim 72\%$ of the reactor volume, while the fastest flow rate yielded a first breakthrough at $\sim 88\%$. The apparent shift in data points in-between 20 and 55 ml/min corresponds to the same flowrates and Dean number where the shift in HETP occurred shown in Figure 1 for the 0 g/L Dextrose data set.

This $V_{\min_apparent}$ can be converted to $T_{\min_apparent}$ by dividing $V_{\min_apparent}$ by the flow rate (Figure 2b). As expected, as the flowrate increased, the time in which the first breakthrough of conductivity occurred decreased. It is noteworthy that the shift seen in Figure 2a is not seen in Figure 2b. This lends itself to the conclusion that the influence of average residence time (i.e., volume of reactor and flow rate) is more influential than dispersive effects to $T_{\min}$ when flow rates are changing. Essentially, when the flow rate decreases, $T_{\min}$ will increase regardless of the dispersive affects.

3.2 | Results of the Φ X-174 pulse injections

Figures 3a,b and 4a show the results of the pulse chase experiments using Φ X-174 along with a T_{Ave} data point for frame of reference. As a reminder, T_{Ave} is the mean residence time, which is calculated by Equation (3) and represents a theoretical reactor that experiences no dispersion. For the Φ X-174 data points, the top error bar represents the first grab sample to be Φ X-174 positive, the bottom error bar represents the last sample to be Φ X-174 negative, and the middle represents the average of these two values. The time difference between T_{Ave} and Φ X-174 first breakthrough quantifies the dispersion that occurs within the

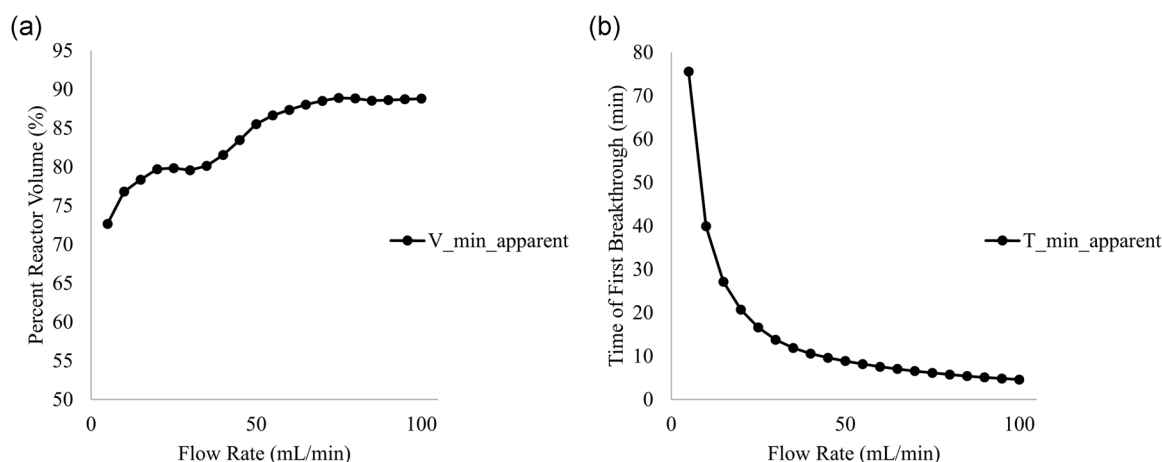


FIGURE 2 Resulting first nonzero conductivity reading, on a (a) volumetric basis and (b) time basis, when a pulse injection of 150 mM NaCl was introduced to the 0.635 cm i.d. JIB and chased with high purity water at a constant flow rate. JIB, Jig-in-a-Box

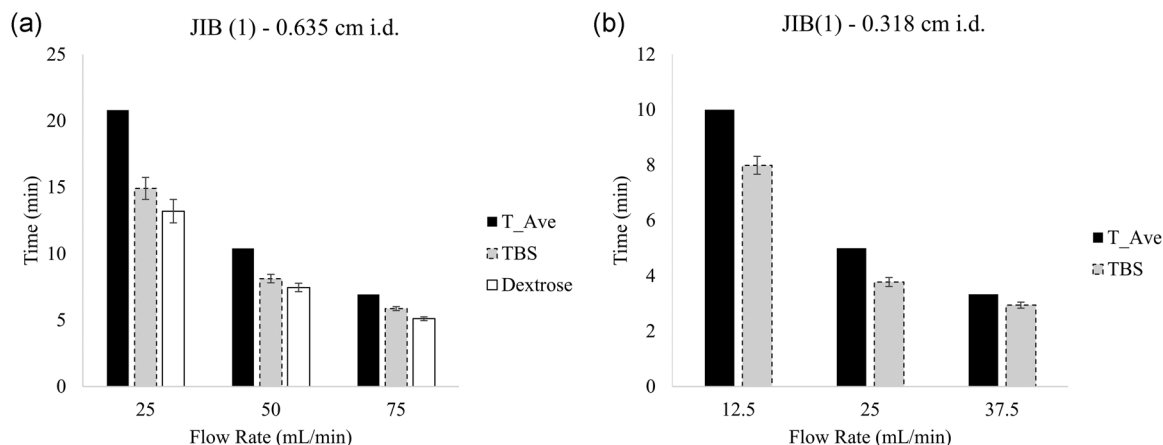


FIGURE 3 Results of the pulse injection experiments with Φ X-174 for one (a) 0.635 cm and (b) 0.318 cm i.d. JIB. The top error bar represents the first grab sample to be Φ X-174 positive, the bottom error bar represents the last sample to be Φ X-174 negative, and the middle represents the average of these two values. JIB, Jig-in-a-Box

reactor. Figure 3a,b displays the results of the two internal diameter JIBs. This Φ X-174 data supports the data generated by the NaCl pulse injection experiments outlined above. This supports the conclusion that the influence of changing flow rates is more impactful than dispersive effects to $T_{\min}$. As the flow rate increases, the Dean number also increases corresponding to an increase in the chaotic nature of Dean vortices that ultimately decreases dispersion resulting in the first breakthrough of Φ X-174 occurring closer to the T_{Ave} value.

When looking at the dextrose data points (Figure 3a), which is representative of a lower Dean number relative to TBS, the instance of first break through is faster than the lower viscosity counterpart. When six of these 0.635 cm i.d. reactors are connected in series (Figure 4a,b) a similar result occurs. Though the bar graph appears to show that the low and high viscosity experiments are nearly identical

(Figure 4a), when analyzing the Φ X-174 titering results versus percent volume of reactor (Figure 4b), the dextrose experiment experiences a slightly earlier breakthrough (i.e., ~1 min) at a higher concentration of Φ X-174 concentration (i.e., 47 pfu vs. 2).

3.3 | Using Dean number to calculate minimum residence time

Figure 5a and b displays a parity plot comparing the 0.318 and 0.635 cm data set, respectively, to the empirical model outlined in Brown et al. (2020). Data points that fell within the Dean number ≥ 100 were aligned with their calculated values, while data points with a Dean number ≤ 100 experienced a later breakthrough than calculated. This was anticipated as the empirical model from Brown

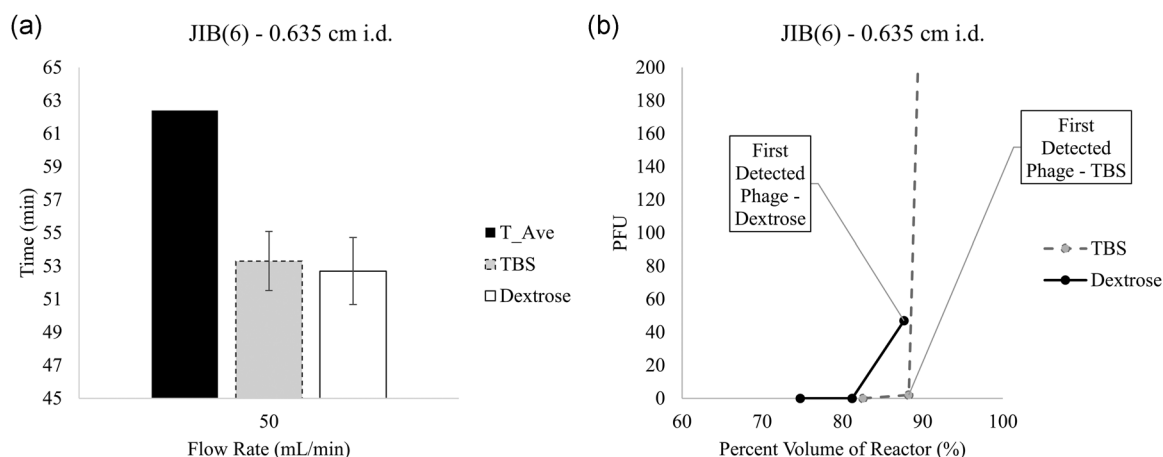


FIGURE 4 (a) Results of the pulse injection experiments with Φ X-174 for the 0.635 cm i.d. with six JIBs connected in series. The top error bar represents the first grab sample to be Φ X-174 positive, the bottom error bar represents the last sample to be Φ X-174 negative, and the middle represents the average of these two values. (b) The titering results of this experiment show an earlier Φ X-174 $T_{\min}$ for the high viscosity solutions. JIB, Jig-in-a-Box; TBS, Tris-buffered saline

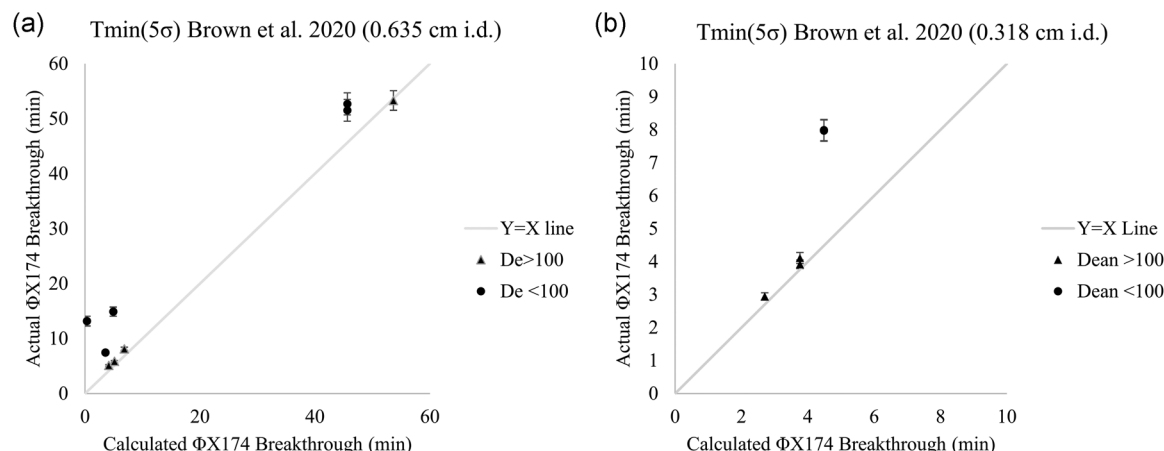


FIGURE 5 A parity plot comparing the breakthrough of Φ X-174 compared to the calculated value generated by the empirical model from Brown et al. (2020) in the (a) 0.635 cm i.d. and (b) 0.318 cm i.d. JIBs. JIB, Jig-in-a-Box

et al. (2020) had incorporated a constraint that the Dean number must be ≥ 100 since the lower Dean number conditions did not follow uniform behavior (Figure 1).

A high-level view of Figure 3a,b data sets reveal an opportunity to fit either a power regression or a third-order polynomial for the prediction of $T_{\min}$. Retrospectively looking back at the data set from Brown et al. (2020), it became apparent that the source of the shift in HETP (Figure 1) was largely impacted by the tailing end of the pulse injection. The asymmetry of the HETP pulses across flow rates ranged from 1.4 to 2.3. Given that an asymmetry of >1 represents a tailing peak, the tailing edge increased the peak's overall variance and therefore affected the HETP and subsequent $T_{\min}$ calculations. Since the front of the peak is the viral safety concern, the tailing portion of the peak was partitioned out of consideration by altering the residence time distribution profiles. The modified trace was created by determining the peak's maximum absorbance and

mirroring the left side trace to the right side of the peak maximum. This was accomplished by exporting the absorbance trace to Microsoft® Excel®, duplicating and inverting the absorbance data set ranging from the first data point until peak max. Data points after peak max were deleted, replaced with the new data points, and each absorbance value was paired with a new time point based on the time difference from peak max. Figure 6a displays a comparison of the peak generated from the detector of the Akta as the dye pulse injection exits the JIB and a modified peak and Figure 6b shows the resulting HETP analysis versus the original method.

Applying this new RTD analysis, recalculating the empirical model using a third-order polynomial and a power regression, Figure 7a–d displays the parity plot comparing the two methods to the Φ X-174 breakthrough. This method allows for the more accurate calculation of first breakthrough for all Dean numbers tested relative to the model in Figure 5. It should be noted that this modified

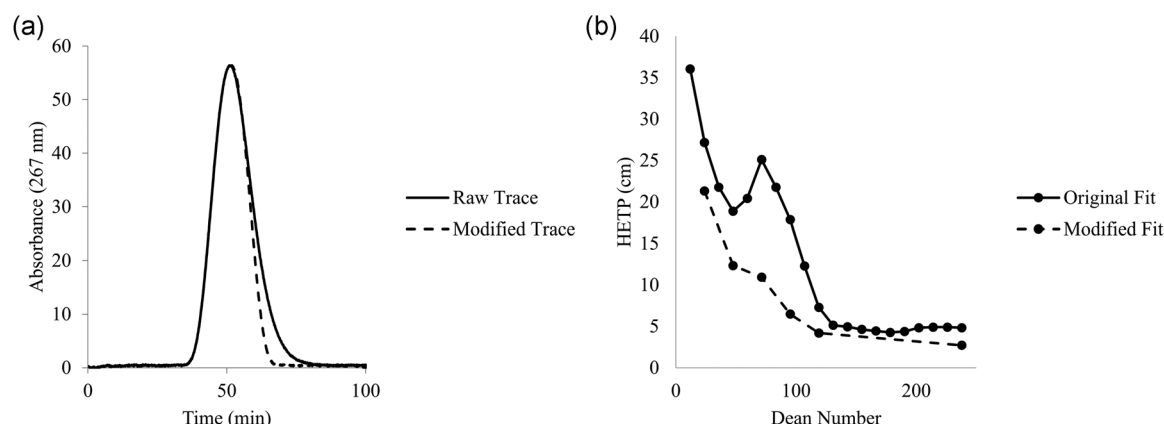


FIGURE 6 (a) Comparison of the trace resulting from the pulse chase experiment (i.e., raw trace) and the modified trace that was created by determining the peak's maximum absorbance and mirroring the left side trace to the right side of the peak maximum to partition out the tailing portion of the peak. (b) Comparison of the HETP data set generated using the raw traced (i.e., original fit) and the modified trace (i.e., modified fit). HETP, height equivalent to a theoretical plate

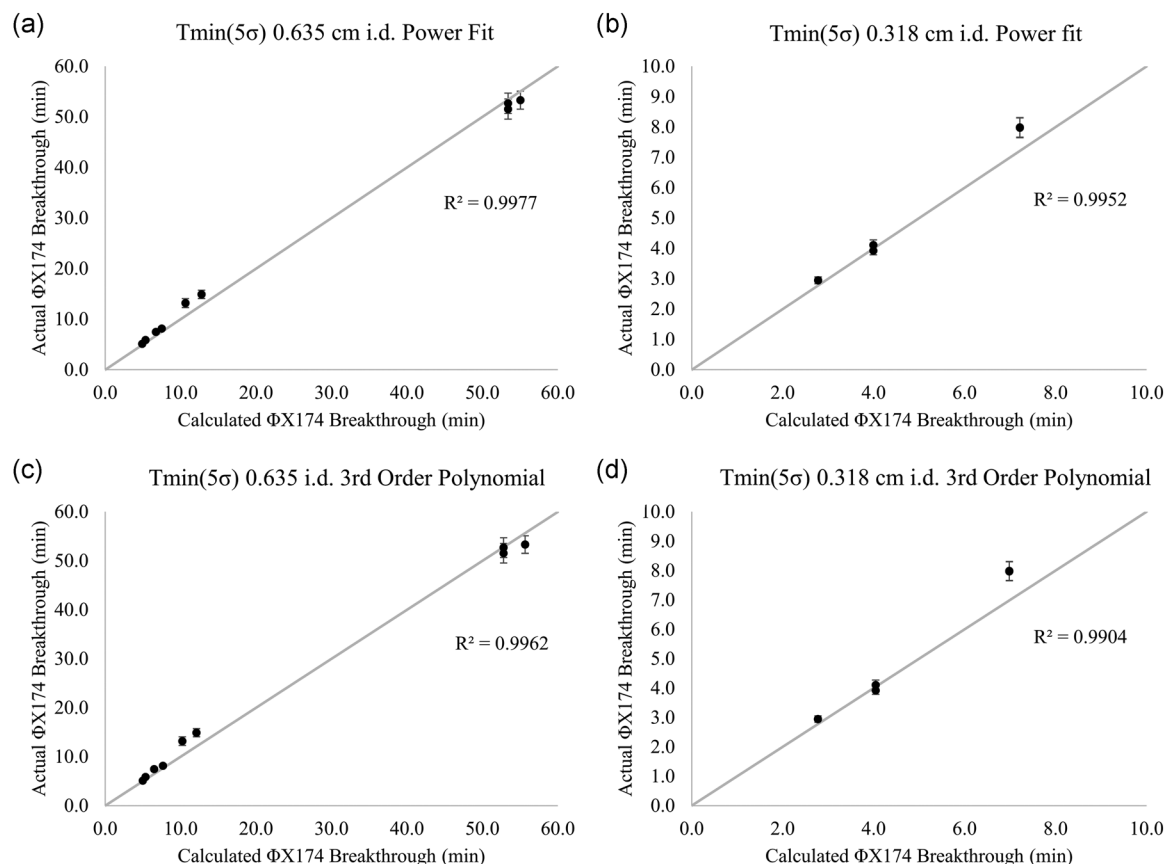


FIGURE 7 Parity plots comparing the calculated to the actual ΦX-174 breakthrough for the power fit ([a] 0.635 cm i.d. and [b] 0.318 cm i.d.) and the third-order polynomial fit ([c] 0.635 cm i.d. and [d] 0.318 cm i.d.)

calculation empirical model ignores the tailing behavior of the RTD and therefore will underestimate the maximum residence time, or time in which the last significant portion of the RTD exits the reactor.

4 | CONCLUSIONS

The primary objective of this study was to understand how viral particles travel through a PFR as a function of reactor i.d., Dean number, and reactor volume, specifically when $T_{\min}$ would occur. The results of the study showed that an increased viscosity can cause earlier breakthrough of viral particles, however the magnitude of the difference was relatively small. When observing a decreasing flow rate, the time in which $T_{\min}$ occurred increased. This leads itself to the conclusion that changing flow rates within the reactor, is more influential than dispersive effects.

As for the ability to calculate first breakthrough of viral particles $T_{\min}(5\sigma)$ using the Brown et al. (2020) model proved to be an effective means of determining the actual $T_{\min}$, but only for the Dean's number ≥ 100 . Through partitioning out the tailing section of the RTD, effective calculation of $T_{\min}$ at all Dean's number conditions was possible with the caveat that the understanding of maximum residence time would be questionable. However, this gap can be

mitigated with a simple pulse injection experiment. With the results of this testing series with ΦX-174, a subset of this study design can now be transferred to the proper model virus (i.e., mammalian virus X-MuLV), and leveraged in the understanding of inactivation kinetics of mammalian viruses within the reactor.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

Matthew R. Brown: *Conceptualization, methodology, investigation, data analysis, data curation, visualization, and writing.* Raquel Orozco: *Project administration and data analysis.*

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